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Project: Inferential Statistics; Hypothesis Testing

Topic 1: Inferential Statistics: Population Mean: One Sample

Dataset: Adult Female Pulse Rates (beats per minute)

Source:

<https://www.statcrunch.com/app/?dlim=comma&ft=true&dataurl=https%3a%2f%2fadx.acs.pearson.com%2fGetPlayerFile.ashx%3fguid%3d90efa010-8428-4fff-bb28-b2985b945041>

Description

The dataset is a sample of the pulse rates of 40 adult females in beats per minute.

Females
69
86
77
80
66
64
78
58
63
84
84
71
76
84
91
86
90
87
96
68
88
79
77
74
54
97
73
75
79

Sample size, $n = 40$

Objectives:

- (1.) Construct a 90% confidence interval estimate of the population mean of the pulse rates of adult females.
- (2.) Interpret the confidence interval.

Parameter to estimate: Population Mean

Test: t-test

Reason for the Test: We were not given the population standard deviation.

Verify Requirements for the Test.

- (1.) The sample is a simple random sample.
- (2.) The sample size is at least 30, justifying normality based on the Central Limit Theorem.

Data Analysis Using R Language

```
# Load the data
data <- read.csv("pulseRatesFemales.csv")

# Method 1: Using the built-in t.test function
confintresult <- t.test(data$Females, conf.level = 0.90)

# Method 2: Formulas
sampleMean <- mean(data$Females)
sampleStandardDeviation <- sd(data$Females)
sampleSize <- length(data$Females)
marginError <-
  qt(0.95, df = sampleSize - 1) * (sampleStandardDeviation / sqrt(sampleSize))

lowerLimit <- sampleMean - marginError
upperLimit <- sampleMean + marginError

cat("90% Confidence Interval:", lowerLimit, "to", upperLimit)
```

90% Confidence Interval: 72.38576 to 79.36424

Interpretation

I am 90% confident that the population mean of Females Pulse Rate (measured in bpm) is approximately between 72.4 bpm and 79.4 bpm (rounded to 1 decimal place).

Topic 2: Hypothesis Testing: Testing a Claim About Population Mean

Dataset: Volumes of Coke cans (in ounces)

Source:

https://www.statcrunch.com/app/?dataurl=www.statcrunch.com%2Fbooksets%2Ftriola_stat14t%2Fdataset%2F37_Cola_Weights_and_Volumes.txt&dlim=tab&ft=true

Description: The dataset is a list of 36 volumes of regular Coke can's (in ounces).

**MyStatLab Data Set**

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Row	Volumes
1	12.31
2	12.14
3	12.19
4	12.33
5	12.24
6	12.3
7	12
8	12.14
9	12.18
10	12.08
11	12.32
12	12.34
13	11.83
14	12.27
15	12.1
16	12.07
17	12.02
18	12.2
19	12.17
20	12.23
21	12.18
22	12.16
23	12.2
24	12.38
25	12.24
26	12.15
27	12.26
28	12.16
29	12.21
30	12.29
31	12.16
32	12.23
33	12.14
34	12.43
35	12.15
36	12.24

Variable: Volume of Coke cans (in ounces)

The accompanying data set includes volumes (ounces) of a sample of cans of regular Coke. The summary statistics are

$$n = 36$$

$$\bar{x} = 12.196 \text{ oz}$$

$$s = 0.114 \text{ oz}$$

Assume that a simple random sample has been selected. Use a 0.01 significance level to test the claim that cans of Coke have a mean volume of 12.00 ounces.

Does it appear that consumers are being cheated?

Objectives:

- (1.) Test the claim that cans of Coke have a mean volume of 12.00 ounces using the Critical Value Method.
- (2.) Test the claim that cans of Coke have a mean volume of 12.00 ounces using the Probability Value Method.
- (3.) State the decision.
- (4.) State the conclusion.
- (5.) State the interpretation.

Verify Requirements for the Test.

- (1.) The sample is a simple random sample.
- (2.) The sample size is at least 30, justifying normality based on the Central Limit Theorem.

Null hypothesis: $\mu = 12 \text{ oz}$

Alternative Hypothesis: $\mu \neq 12 \text{ oz}$

Test: One sample: Two- Tailed

Data Analysis Using R Language:

```

▶ # Load the data
data <- read.csv("Volumes.csv")
volumes <- data$Volumes

# Parameters
PopulationMean <- 12.00
alpha <- 0.01
SampleSize <- length(volumes)
df <- SampleSize - 1

# Calculate Sample Statistics
MeanValue <- mean(volumes)
sdVal <- sd(volumes)
stanardError <- sdVal / sqrt(SampleSize)

# 1. P-Value Method & Confidence Interval Method
t_test_results <- t.test(volumes, mu = PopulationMean, ConfidenceValue = 0.99)
print(t_test_results)

# 2. Critical Value Method
TestStatistic <- (MeanValue - PopulationMean) / se
TestCritical <- qt(1 - alpha/2, df = df)

cat("T-statistic:", TestStatistic, "\n")
cat("Critical Value:", TestCritical, "\n")

```

...

One Sample t-test

```

data: volumes
t = 10.303, df = 35, p-value = 3.856e-12
alternative hypothesis: true mean is not equal to 12
95 percent confidence interval:
 12.15702 12.23409
sample estimates:
mean of x
 12.19556

T-statistic: 10.3034
Critical Value: 2.723806

```

Critical Value Method

$$\text{Critical } z - \text{value} = 2.723806$$

$$\text{Test Statistic} = 10.3034$$

$$10.3034 > 2.723806$$

The Test Statistic is greater than the Critical z.

It falls in the critical region.

The decision is to reject the null hypothesis.

P-Value Method

$$P - \text{Value} = 0$$

$$\text{Level of Significance} = 0.01$$

$$0 < 0.01$$

P-Value is less than level of significance.

The decision is to reject the null hypothesis.

Conclusion

There is sufficient evidence to warrant the rejection of the claim that cans of Coke have a mean volume of 12.00 ounces.

Interpretation

The population mean volume of Coke cans is significantly different from the stated value of 12 ounces.

References (MLA 9)

“StatCrunch.” *Statcrunch.com*, 2026,

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iola_stat14t%2Fdataset%2F37_Cola_Weights_and_Volumes.txt&dlim=tab
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ftdx.acs.pearson.com%2fGetPlayerFile.ashx%3fguid%3d90efa010-8428-
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